THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 11: **Areas Related to Circles** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An arc of length 44 cm subtends an angle at the centre of a circle of radius 14 cm. The angle subtended by the arc at the centre is: 1

(a) 210° **(b)** 360° **(c)** 180° **(d)** 90°

2. A chord of a circle of radius 63 cm subtends a right angle at the centre. The area of the minor segment is: 1

(a) 2126.25 cm² **(b)** 1984.5 cm² **(c)** 1134 cm² **(d)** 3118.5 cm²

3. A wire in the form of a circle of radius 49 cm is bent into a square. The area of the square is: 1

(a) 23716 cm² **(b)** 77 cm² **(c)** 7546 cm² **(d)** 5929 cm²

4. A quadrant of a circle of radius 35 cm is cut from a square of side 35 cm, with the centre of the quadrant at one corner of the square. The area of the remaining part of the square is: 1

(a) 962.5 cm² **(b)** 1225 cm² **(c)** 262.5 cm² **(d)** 525 cm²

5. A square is inscribed in a circle of radius 28 cm. The area of the circle outside the square is: 1

(a) 896 cm² **(b)** 2464 cm² **(c)** 1680 cm² **(d)** 1568 cm²

6. The largest possible circle is cut from a square sheet of side 63 cm. The area of the sheet left over is: 1

(a) 1701 cm² **(b)** 850.5 cm² **(c)** 3969 cm² **(d)** 3118.5 cm²

7. A circular park of radius 40 m is surrounded by a path 7 m wide. The area of the path is: 1

(a) 1760 m² **(b)** 957 m² **(c)** 3828 m² **(d)** 1914 m²

8. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 30° is marked in both. The area of the region between the two arcs within the sector is: 1

(a) 77 cm² **(b)** 462 cm² **(c)** 19.25 cm² **(d)** 38.5 cm²

9. An arc of length 33 cm subtends an angle at the centre of a circle of radius 14 cm. The angle subtended by the arc at the centre is: 1

(a) 45° **(b)** 270° **(c)** 135° **(d)** 145°

10. A chord of a circle of radius 70 cm subtends a right angle at the centre. The area of the minor segment is: 1

(a) 2625 cm² **(b)** 1400 cm² **(c)** 2450 cm² **(d)** 3850 cm²

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. A wheel has radius 7 cm. Find the number of complete revolutions it makes to cover 22 m. 2

12. The minute hand of a clock is 28 cm long. Find the area swept by the minute hand in 9 minutes. 2

13. A horse is tied at the midpoint of one side of a large square field with a rope 7 m long. Find the area of the field it can graze. 2

14. A wheel has radius 28 cm. Find the number of complete revolutions it makes to cover 88 m. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A circular field has radius 42 m. Fencing costs ₹ 7 per metre. Find the cost (in ₹) of fencing the field once. 3

16. The radii of two circles are 21 cm and 16 cm. Find the radius of the circle whose circumference equals the sum of their circumferences. 3

17. The radii of two circles are 10 cm and 24 cm. Find the radius of the circle whose area equals the sum of their areas. 3

OR

A circular field has radius 49 m. Fencing costs ₹ 5 per metre. Find the cost (in ₹) of fencing the field once. 3

18. The radii of two circles are 9 cm and 4 cm. Find the radius of the circle whose circumference equals the sum of their circumferences. 3

OR

The radii of two circles are 18 cm and 24 cm. Find the radius of the circle whose area equals the sum of their areas. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 36° is marked in both. Find the area of the region between the two arcs within the sector. 5

OR

A horse is tied at a corner of a large square field. The rope is first 14 m long and then lengthened to 35 m. Find the increase in the area it can graze. 5

20. Two concentric circles have radii 14 cm and 7 cm. A sector of angle 24° is marked in both. Find the area of the region between the two arcs within the sector. 5

OR

A horse is tied at a corner of a large square field. The rope is first 21 m long and then lengthened to 42 m. Find the increase in the area it can graze. 5

END OF QUESTION PAPER — Total: 40 Marks